

Dietary Effects of Menhaden Oil on the Growth and Membrane Lipid Composition of Rat Mammary Tumors^{1,2,3,4}

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ABSTRACT—The objective of this study was to evaluate the effect of quantitative differences in dietary menhaden oil, an *n*-3 polyunsaturated marine oil, on mammary tumor development and to compare these results with those produced by corn oil, an *n*-6 polyunsaturated vegetable oil. Inbred female BUF rats were treated with the carcinogen *N*-methyl-*N*-nitrosourea (CAS: 684-93-5) and maintained on diets containing graded levels of each of the respective oils. Our results indicated that diets containing 20% menhaden oil produced a reduction in tumor incidence and a prolongation of the tumor latent period. This finding contrasted sharply with the enhanced tumor development and shortened latent period observed in the animals fed the equivalent dietary level of corn oil. Fatty acid analyses performed on the lipids extracted from the tumor and hepatic microsomes of the animals on the menhaden oil diets demonstrated that the proportion of eicosapentaenoic acid (No. of C-atoms:No. of double bonds = 20:5) present in these microsomal lipids was related inversely to mammary tumor development.—JNCI 1985; 74:1145-1150.

The effects of dietary lipids on mammary tumor development are complex and incompletely understood. It has been repeatedly demonstrated that quantitative increases in the dietary content of a polyunsaturated vegetable oil, such as corn oil, can enhance mammary tumor development in rats, but the biochemical mechanisms responsible for this phenomenon are unclear (2-9). It is well known that quantitative and qualitative differences in dietary lipid can significantly influence the fatty acid composition of cell membranes, and it has even been shown that some of these diet-induced changes can alter the lactogenic hormone-binding capacities of mammary tumor membranes (10). Such changes in lactogenic hormone-binding capacity alone, however, are unlikely to account for all of the dietary effects of lipids on mammary tumor development because it is now recognized that these induced alterations in hormone binding are restricted normally to conditions where the dietary polyunsaturated lipid level is maintained below a threshold of approximately 3% (11, 12). It is evident, therefore, that other biochemical processes must be involved.

In this regard, attention has been focused recently on the possibility that dietary lipids can influence tumor development through their effects on prostaglandin metabolism (13, 14). It is known that certain polyunsaturated fatty acids that are found in high levels in vegetable oils, such as linoleic acid (C18:2),⁸ can be converted to arachidonic acid (C20:4) and thus serve as substrates for prostaglandin synthesis. Consequently, it has been postulated that diets containing high levels of these *n*-6 fatty acids (i.e., those having their first double

methyl end of the molecule) could affect tumor development by enhancing prostaglandin synthesis.

It has also been observed that certain *n*-3 polyunsaturated fatty acids (i.e., those having their first double bond between the third and fourth carbons from the methyl end of the molecule), such as eicosapentaenoic acid (C20:5), behave as competitive inhibitors of prostaglandin endoperoxide synthetase (15-17). Since these *n*-3 lipids are present in high levels of marine oils, such as menhaden oil, this study sought to evaluate how quantitative variations in the dietary content of menhaden oil would affect mammary tumor development and how these effects would compare with those produced by equivalent dietary variations in corn oil.

MATERIALS AND METHODS

Treatment groups.—Experiment 1: Six-week-old inbred female BUF rats (Simonsen Laboratories, Gilroy, CA) were divided into 3 diet groups. These diets contained either 0.5, 3, or 20% menhaden oil, respectively. The exact compositions of these diets are presented in table 1. The formulas for these semisynthetic diets were purposely chosen to follow those of previously established experimental diets (10, 11). They provided an equivalent ratio of protein calories to total calories, and the vitamin and mineral contents of the diets were

ABBREVIATION USED: MNU = *N*-methyl-*N*-nitrosourea.

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⁸ Parenthetical values are the No. of C-atoms:No. of double bonds.

TABLE 1.—*Diet composition*

Component	Diet		
	0.5% lipid	3% lipid	20% lipid
Corn oil or menhaden oil	5 g/kg	30 g/kg	200 g/kg
Casein (vitamin-free)	180 g/kg	190 g/kg	230 g/kg
Sucrose	220 g/kg	205 g/kg	145 g/kg
Dextrose	260 g/kg	240 g/kg	175 g/kg
Dextrin	260 g/kg	240 g/kg	175 g/kg
Rogers Harper vitamin mix	5 g/kg	5 g/kg	5 g/kg
Rogers Harper salt mix	50 g/kg	50 g/kg	50 g/kg
Vitamin B ₁₂ (10 µg/ml water)	5 ml/kg	5 ml/kg	5 ml/kg
Choline chloride (0.20 g/ml water)	15 ml/kg	15 ml/kg	15 ml/kg

sufficient to meet all growth requirements. Specific care was taken to maintain the menhaden oil in the dark at 4°C until used in preparation of the diets. New diet was prepared every 7–10 days, and it was stored under similar conditions as those used for the oil itself. Fresh diet was distributed to the animals every 24–48 hours.

Experiment 2: Six-week-old female BUF rats were divided into 5 diet groups. These diets contained either 0.5 or 20% menhaden oil, 0.5 or 20% corn oil, or Formulab Chow 5008 (Ralston Purina Co., St. Louis, MO), a standard commercially available rat diet. The exact compositions of the four semisynthetic diets are presented in table 1. The corn oil and choline used in these diets were purchased from ICN Nutritional Biochemicals Corp., Cleveland, OH. All other ingredients except the menhaden oil came from Teklad, Inc., Madison, WI. The menhaden oil, whose fatty acid profile was previously reported by Culp et al. (18), was provided without cost by Mr. A. P. Bimbo, Research Director, Zapata Haynie Corp., Reedville, VA.

Tumor induction.—At 50 days of age, all animals received the carcinogen MNU (CAS: 684-93-5; ICN Life Sciences Group, K & K Laboratories, Plainview, NY). Following the method of Gullino et al. (19), the MNU was dissolved in water (10 mg/ml) and was administered iv at a dose of 5 mg/100 g body weight. Similar doses were subsequently administered to the rats at 78 and 106 days of age. All animals were maintained under identical environmental conditions, and the development of the mammary tumors was monitored weekly. When each rat's tumor reached approximately 1–2 cm in diameter, the unanesthetized animal was killed by cervical transection early in the day during diestrus. Immediately afterward, an autopsy was performed, and samples of the tumor and liver were snap frozen in liquid nitrogen and stored at –80°C.

Subcellular fractionation.—The frozen tissue samples were homogenized in a tissuemizer (Tekmar Co., Cincinnati, OH) and then differentially separated into specific subcellular fractions by centrifugation with the use of previously described methods (11). The resultant microsome-membrane fraction was then saved for lipid

analysis by storage at –80°C in a gaseous nitrogen atmosphere.

Qualitative fatty acid analyses.—The lipids in the microsome-membrane fractions were extracted according to the method of Folch et al. (20). They were subsequently methylated and stored by means of techniques that we have described previously (11). At the time of final analysis, these methylated lipids were dried down under gaseous nitrogen and redissolved in 25–50 µl of *n*-hexane. Then 2 µl of this solution was injected into a gas chromatograph (Hewlett-Packard 5840-A) containing a 6-foot glass column packed with 10% SP-2330 on a 100/120 chromosorb W acid-washed support (Supelco, Inc., Bellefonte, PA). The retention times of the individual fatty acids and the integrated areas under each peak were obtained directly from the computer associated with the instrument. We determined the identity of the observed peaks by comparing them with known standards (Supelco, Inc., and Sigma Chemical Co., St. Louis, MO) run at the time of assay.

Data analyses.—All results represent the values of an entire group and are reported as the mean plus or minus the standard error of the mean. The data analyses were performed by the Department of Biostatistics, University of Rochester School of Medicine, with the use of the forms of statistical analysis listed below each of the tables.

RESULTS

Experiment 1

Autopsy Data

The autopsy results are presented in table 2. They showed that as the percentage of menhaden oil in the diet increased, there was a progressive lengthening of the tumor latent period as well as a reduction in tumor incidence. These differences became highly significant in the 20% menhaden oil group. There was no evidence of any significant differences in body weights among the 3 diet groups, and there were no observable signs of malnutrition or dietary deficiency.

Mammary Tumor Fatty Acid Profiles

The fatty acid profiles produced from the lipids extracted from the tumor microsomes are presented in table 3. These results indicated that the quantitative differences in the dietary content of menhaden oil produced relatively selective alterations in these profiles. There were, for example, clearly defined increases in the proportions of eicosapentaenoic acid (C20:5), docosapentaenoic acid [C22:5 (*n*-3)],⁹ and docosahexaenoic acid (C22:6) as the dietary content of menhaden oil increased, but there were essentially no differences in the proportions of myristic acid (C14:0), palmitic acid (C16:0), palmitoleic acid (C16:1), stearic acid (C18:0), oleic acid

⁹*n* = the location of the first double bond from the methyl end of the fatty acid.

TABLE 2.—Autopsy data^a

Diet group	No.	Age at death, days	Body weight, g	Tumor incidence, %	Tumor No.	Tumor burden, g
Expt 1						
Menhaden oil, 0.5%	9	146.4±4.3	205.6±4.5	100	3.0±0.4	8.5±0.8
Menhaden oil, 3%	10	156.3±5.0	219.4±6.2	100	2.9±0.3	8.1±1.2
Menhaden oil, 20%	8	171.1±5.7 ^{b,c}	207.0±9.5	62.5 ^{d,e}	1.6±0.5	4.3±1.8
Expt 2						
Menhaden oil, 0.5%	11	156.7±4.9	202.4±3.5	100	3.5±0.4	8.8±1.2
Corn oil, 0.5%	10	167.3±8.2	203.9±2.8	100	3.1±0.3	7.4±1.1
Rat chow	11	158.3±6.0	220.6±3.2 ^{b,f,g,h}	100	3.5±0.4	10.1±1.4
Menhaden oil, 20%	8	219.1±11.5 ^{b,f,h,i}	207.1±5.3	62.5 ^j	0.88±0.13 ^{b,f,h,i}	2.04±0.76 ^{b,f,h,i}
Corn oil, 20%	9	151.2±5.0	204.2±4.8	100	2.9±0.3	7.0±1.4

^a Values are means ± SE; data are from all animals in each group.

^b Difference from 0.5% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.

^c Difference from 3% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.

^d Difference from 0.5% menhaden oil group significant at $P < .05$ by chi-square analysis.

^e Difference from 3% menhaden oil group significant at $P < .05$ by chi-square analysis.

^f Difference from 0.5% corn oil group significant at $P < .05$ by Newman-Keuls analysis.

^g Difference from 20% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.

^h Difference from 20% corn oil group significant at $P < .05$ by Newman-Keuls analysis.

ⁱ Difference from rat chow group significant at $P < .05$ by Newman-Keuls analysis.

^j The probability that this reduction in tumor incidence would occur only in the 0.5% menhaden oil group is $< .05$.

(C18:1), docosatetraenoic acid (C22:4), and docosapentaenoic acid [C22:5 (*n*-6)]. Slight decreases in the proportions of linoleic acid (C18:2) and arachidonic acid (C20:4) also did occur in the lipids of the 20% menhaden oil group; however, the absence of a tight correlation between the dietary levels of menhaden oil in the other 2 groups and the proportions of these fatty acids made the significance of these changes somewhat equivocal.

Hepatic Fatty Acid Profiles

The fatty acid profiles of the hepatic microsomes are presented in table 4. These results again demonstrated the selective nature of the changes in the microsome lipids induced by the diets and indicated that, despite some organ-specific differences, the fatty acid profiles of the hepatic lipids nicely highlighted many of the features observed in the tumor lipids. The positive correlation between the amount of menhaden oil in the diet and proportions of eicosapentaenoic acid (C20:5) and docosapentaenoic acid [C22:5 (*n*-3)] was still very evident, and this correlation was even more prominent for docosahexaenoic acid (C22:6) in the hepatic microsomes than it had been in the tumor. In the hepatic microsomes, however, there was also an unequivocal negative correlation between the amount of menhaden oil in the diet and the proportion of oleic acid (C18:1), which had not been noted in the tumor microsomes. Slight decreases in arachidonic acid (C20:4) and docosatetraenoic acid (C22:4) could also be seen in the microsomes of the 20% menhaden oil group.

Experiment 2

Autopsy Data

The autopsy findings of the animals used in this experiment are presented in table 2. As observed in

experiment 1, these data again confirmed that the 20% menhaden oil group had a definitely reduced mammary tumor incidence and a prolonged latent period. Likewise, the 0.5% corn oil group demonstrated a prominent delay in tumor latency, although this was not quite as statistically significant as in previous studies (10, 11). In sharp contrast, the 20% corn oil group displayed a markedly shortened tumor latent period indicative of enhanced tumor development.

Mammary Tumor Fatty Acid Profiles

The fatty acid composition of the lipids extracted from the tumor microsomes prepared in experiment 2 is presented in table 3. As in experiment 1, these results showed that the individuality of the fatty acid profiles of each of the diet groups was primarily dependent on differences in certain specific fatty acids. The values for palmitoleic acid (C16:1) were elevated in both of the fish oil groups, reduced in the rat chow group, and intermediate in both of the corn oil groups. Oleic acid (C18:1) was highest in the 0.5% menhaden oil group and lowest in the 20% corn oil group, with the values for the 0.5% menhaden oil, 0.5% corn oil, and 20% menhaden oil groups being roughly equivalent. Linoleic acid (C18:2) values were significantly higher in the 20% corn oil group and in the rat chow group than in any of the remaining groups. The values for arachidonic acid (C20:4) and docosatetraenoic acid (C22:4) were significantly higher in both of the corn oil groups and in the rat chow group than in either of the menhaden oil groups, but there was no simple correlation between the specific quantitative amount of either dietary oil and the proportion of these fatty acids in the microsome lipids. As in experiment 1, the proportion of eicosapentaenoic acid (C20:5) was noted to be greatly increased in the 20% menhaden oil group, and the results for docosapentaenoic acid [C22:5 (*n*-3)] were noted to follow a similar pattern.

TABLE 3.—Fatty acid composition of lipids extracted from tumor microsomes^a

Diet group	No.	Fatty acid ^b											
		C14:0	C16:0	C16:1	C18:0	C18:1	C18:2	C20:4	C20:5	C22:4	C22:5 (n-6)	C22:5 (n-3)	C22:6
Expt 1													
Menhaden oil, 0.5%	9	1.8±0.7	18.0±1.0	1.8±0.2	11.0±0.3	25.7±1.1	1.5±0.2	12.7±0.9	2.7±0.3	4.9±0.4	5.3±0.4	1.9±0.2	4.4±0.3
Menhaden oil, 3%	10	1.2±0.6	19.6±1.1	1.9±0.3	12.1±0.5	26.8±1.2	1.8±0.1	9.2±0.5 ^c	5.4±0.6 ^c	3.6±0.6	3.6±0.7	3.3±0.3 ^c	4.8±0.2
Menhaden oil, 20%	5	1.5±0.8	19.4±1.1	1.7±0.4	11.9±0.7	22.8±1.6	1.0±0.1 ^{c,d}	10.1±0.9 ^c	7.5±1.0 ^{c,d}	3.6±0.2	4.1±0.2	4.1±0.4 ^{c,d}	5.8±0.3 ^c
Expt 2													
Menhaden oil, 0.5%	6	6.3±0.6	18.3±0.3	3.4±0.3	12.5±0.1	22.1±0.8	1.5±0.1	10.1±0.5	2.1±0.2	2.5±0.2	3.8±0.4	1.5±0.2	3.2±0.1
Corn oil, 0.5%	5	4.2±1.2	17.7±0.6	2.3±0.4	13.8±1.8	18.2±0.8 ^c	1.5±0.3	16.3±1.0 ^c	1.3±0.2	6.0±0.8 ^c	3.8±0.4	0.6±0.2 ^c	3.5±1.6
Rat chow	5	3.0±1.2	19.1±0.7	0.9±0.4 ^{c,e}	13.5±0.4	18.2±0.7 ^c	5.4±0.6 ^{c,e}	17.5±1.0 ^c	1.0±0.2	4.5±0.5 ^c	3.0±0.2	1.4±0.1 ^e	2.9±0.1
Menhaden oil, 20%	5	5.2±0.4	19.6±0.6	3.3±0.4 ^f	12.9±0.6	18.7±1.0 ^c	2.0±0.2 ^f	10.1±1.1 ^{c,f}	4.4±0.6 ^{c,e,f}	2.5±0.3 ^{e,f}	3.2±0.4	2.9±0.3 ^{c,e,f}	4.3±0.2
Corn oil, 20%	5	5.8±0.6	19.0±0.2	1.9±0.5 ^c	12.1±0.3	14.8±0.4 ^{c,e,g}	6.4±0.7 ^{c,e,g}	14.7±0.7 ^{c,g}	1.1±0.1 ^g	5.8±0.2 ^{c,g}	2.8±0.1	1.3±0.1 ^{e,g}	1.7±0.1

^a Values are means ± SE; mole percent of total lipid.^b No. of C-atoms: No. of double bonds. Numbers in parentheses indicate the location of the first double bond from the methyl end of the fatty acid.^c Difference from 0.5% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.^d Difference from 3% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.^e Difference from 0.5% corn oil group significant at $P < .05$ by Newman-Keuls analysis.^f Difference from rat chow group significant at $P < .05$ by Newman-Keuls analysis.^g Difference from 20% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.TABLE 4.—Fatty acid composition of lipids extracted from liver microsomes^a

Diet group	No.	Fatty acid ^b											
		C14:0	C16:0	C16:1	C18:0	C18:1	C18:2	C20:4	C20:5	C22:4	C22:5 (n-6)	C22:5 (n-3)	C22:6
Expt 1													
Menhaden oil, 0.5%	5	1.9±0.8	13.7±2.2	2.5±0.6	19.8±0.8	19.5±1.6	2.8±0.3	10.5±0.6	4.6±0.7	1.7±0.2	1.8±0.2	1.0±0.1	14.4±2.0
Menhaden oil, 3%	5	1.8±0.6	17.5±0.6	2.8±0.2	20.1±1.1	15.0±2.4	3.1±0.2	7.5±0.4 ^c	9.1±1.2	1.4±0.1	1.2±0.2	2.0±0.3 ^c	16.2±1.1
Menhaden oil, 20%	5	1.3±0.6	17.3±1.2	2.7±0.4	18.6±0.6	11.0±1.8 ^c	2.5±0.4	9.4±0.4 ^d	8.4±0.6 ^c	0.9±0.2 ^{c,d}	1.1±0.1	2.1±0.4 ^c	22.3±0.7 ^{c,d}
Expt 2													
Menhaden oil, 0.5%	6	0.3±0.1	16.0±0.7	3.1±0.3	20.8±0.8	19.5±1.2	2.4±0.2	9.6±0.7	4.5±0.7	1.5±0.2	1.5±0.2	0.9±0.1	13.6±0.9
Corn oil, 0.5%	5	0.5±0.1	15.2±0.4	3.0±0.3	20.1±0.5	18.6±0.6	3.0±0.2	21.8±0.5 ^c	1.4±0.4 ^c	2.2±0.4	5.4±0.4 ^c	0.5±0.1	3.5±0.2 ^{c,e}
Rat chow	5	0.6±0.2	14.8±0.2	0.3±0.2 ^c	23.2±0.6	8.9±0.2 ^{c,e}	10.4±0.8 ^{c,e}	21.2±0.6 ^c	1.7±0.3	1.5±0.3	1.7±0.9 ^e	1.3±0.2 ^e	11.2±1.4 ^{c,e}
Menhaden oil, 20%	5	1.0±0.3 ^{c,e}	16.2±0.4	3.0±0.4 ^f	20.9±0.2	7.9±0.3 ^{c,e}	3.2±0.3 ^g	10.5±2.5 ^g	11.3±0.7 ^{c,e,g}	1.0±0.1	1.1±0.1 ^e	1.9±0.2 ^{c,e,g}	17.9±0.5 ^{c,e,g}
Corn oil, 20%	5	0.2±0.2 ^f	13.9±0.7	0.2±0.2 ^{c,e,f}	21.5±1.2	9.6±1.0 ^{c,e}	15.8±1.5 ^{c,e,f,g}	22.1±1.2 ^{c,f}	0.5±0.3 ^{c,f}	2.6±0.1	4.0±0.5 ^{c,f,g}	0.6±0.2 ^{f,g}	4.7±0.4 ^{c,e,f,g}

^a Values are means ± SE; mole percent of total lipid.^b No. of C-atoms: No. of double bonds. Numbers in parentheses indicate the location of the first double bond from the methyl end of the fatty acid.^c Difference from 0.5% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.^d Difference from 3% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.^e Difference from 0.5% corn oil group significant at $P < .05$ by Newman-Keuls analysis.^f Difference from rat chow group significant at $P < .05$ by Newman-Keuls analysis.^g Difference from 20% menhaden oil group significant at $P < .05$ by Newman-Keuls analysis.

Hepatic Fatty Acid Profiles

The fatty acid profiles of the hepatic microsomes are presented in table 4. As shown there, the myristic acid (C14:0) values were somewhat higher for the 20% menhaden oil group than the other groups, whereas the palmitoleic acid (C16:1) values were increased in both of the menhaden oil groups as well as the 0.5% corn oil group. Oleic acid (C18:1) values were elevated also in both of the 0.5% lipid groups, but linoleic acid (C18:2) values were elevated only in the 20% corn oil and rat chow groups. Arachidonic acid (C20:4) and docosapentaenoic acid [C22:5 (*n*-6)] values were highest in both of the corn oil groups and the rat chow group, whereas eicosapentaenoic acid (C20:5) and docosahexaenoic acid (C22:6) values were elevated exclusively in the menhaden oil groups.

DISCUSSION

Although earlier studies (21) in which a single dose of menhaden oil in the diet was used had suggested that *n*-3 lipids could influence mammary tumor growth in the same fashion as the *n*-6 polyunsaturated lipids (11, 14, 22, 23), our results with multiple doses of dietary menhaden oil indicate that this is not the case. It was clearly evident that animals maintained on a 20% menhaden oil diet had a significant delay in mammary tumor development, which was progressively less apparent in animals maintained on 3 and 0.5% diets.

Our results do not provide the precise biochemical explanation for the contrasting effects produced by these two types of polyunsaturated lipid, but they do show that modifications in the dietary content of either lipid type significantly influence the fatty acid profiles of both the tumor and hepatic cell membranes. It would appear, therefore, that each type of polyunsaturated lipid has selective effects on the fatty acid composition of the membrane lipids that can be accentuated by a quantitative increase in the dietary content of that lipid.

Previous studies with the use of rats maintained on varying levels of dietary corn oil (10, 11) have shown that mammary tumor development and tumor lactogenic hormone binding can be decreased if the mean levels of desaturation of tumor membrane fatty acids are lowered. These observations have led investigators to postulate that these reductions in lipid desaturation importantly influence mammary tumor cell development by their effects on membrane fluidity. Our present experimental results do not necessarily challenge the validity of this perception; however, they do seem to indicate that dietary lipids can affect tumor development by other biochemical mechanisms as well. The occurrence of reduced mammary tumor development in the 20% menhaden oil group, for example, despite the presence of a high level of desaturation in the tumor membranes, shows that tumor development can be delayed by a dietary mechanism not involving a reduction in the mean level of membrane fatty acid desaturation. Similarly, the observation that the 0.5% menhaden oil group had a greater tumor development than the 0.5% corn oil group, even though

their mean levels of microsome fatty acid desaturation were essentially equivalent, implies that additional biochemical features of the lipids must also be involved. We believe that these observations highlight the importance of the structural differences between these two types of polyunsaturated lipid. It seems very probable that even the physical-chemical properties of a given tumor membrane depend not only on the mean level of desaturation but also on the distinctive locations of the double bonds in the majority of the fatty acids.

Our data also show that progressive increases in the specific membrane fatty acids linoleic acid (C18:2) and eicosapentaenoic acid (C20:5) were associated with quantitative increases in dietary corn oil and fish oil, respectively. To what extent either of these fatty acid variations is causally related to the observed differences in tumor development is uncertain, but the fact that eicosapentaenoic acid has been shown to disturb prostaglandin synthesis makes it highly probable that the increasing proportions of this fatty acid in the membrane are significantly altering the tumor's capacity for intracellular prostaglandin synthesis. The recent studies of Karmali et al. (17) on the R3230AC mammary tumor and of Abraham and Hillyard (24) on mammary adenocarcinomas in mice appear to support this hypothesis. Precisely how such disturbances in prostaglandin metabolism affect tumor growth is presently unclear. Several of the more attractive suggestions include modulation of intracellular hormone action (25), alterations in the physical-chemical characteristics of the membrane (26), and attenuation of host defense mechanisms (27). Certainly, continued investigation into these mechanisms is needed if we are to understand fully how dietary lipids influence breast cancer.

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